Advanced & Specialised AI Topics

Everything else on the road to mastery: causal inference, probabilistic ML, AutoML, time series with DL, privacy, robotics, quantum and more.

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1. Causal inference [Advanced]

Explanation (English)
Correlation is not causation. Use randomised experiments, DAGs, propensity scores, difference-in-differences, instrumental variables, uplift modelling (DoWhy, EconML).

Tanglish (Tamil + English) Vilakkam
Correlation != causation. Causal inference: DAG, propensity score.

Example / Program
```
# import dowhy; model=dowhy.CausalModel(data=df,treatment='t',outcome='y',graph=g)
```

Learn more
YouTube search: causal inference for data science

2. Probabilistic programming and Bayesian ML (PyMC, Stan, Pyro) [Advanced]

Explanation (English)
Specify generative models and infer posteriors with MCMC / variational inference; Gaussian processes; Bayesian neural nets.

Tanglish (Tamil + English) Vilakkam
PyMC-la Bayesian models.

Example / Program
```
import pymc as pm
with pm.Model():
    mu=pm.Normal('mu',0,1);pm.Normal('y',mu,1,observed=[0.2,0.4,0.1]);idata=pm.sample(200)
```

Learn more
YouTube search: Probabilistic programming and Bayesian ML (PyMC, Stan, Pyro) tutorial

3. Deep learning for time series (LSTM, TCN, Transformers, N-BEATS, TimesFM) [Advanced]

Explanation (English)
Windowed supervised learning, multi-horizon forecasting, foundation models for time series; avoid leakage with time-aware splits.

Tanglish (Tamil + English) Vilakkam
LSTM / Transformers time series forecast.

Example / Program
```
import numpy as np
s=np.sin(np.arange(200)/5)
X=np.array([s[i:i+20] for i in range(len(s)-20)]);y=s[20:]
print(X.shape,y.shape)
```

Learn more
YouTube search: Deep learning for time series (LSTM, TCN, Transformers, N-BEATS, TimesFM) tutorial

4. AutoML and neural architecture search [Advanced]

Explanation (English)
Automate preprocessing, model choice and tuning (AutoGluon, FLAML, H2O, auto-sklearn, Optuna).

Tanglish (Tamil + English) Vilakkam
AutoML: automatic model selection.

Example / Program
```
# from autogluon.tabular import TabularPredictor; TabularPredictor('y').fit(train)
```

Learn more
YouTube search: AutoML and neural architecture search tutorial

5. Federated learning, differential privacy, secure ML [Advanced]

Explanation (English)
Train across devices without centralising data; add calibrated noise (DP-SGD); homomorphic encryption.

Tanglish (Tamil + English) Vilakkam
Federated learning, differential privacy.

Learn more
YouTube search: federated learning explained differential privacy

6. Explainable and trustworthy AI [Advanced]

Explanation (English)
Interpretability methods (SHAP, LIME, Grad-CAM, attention analysis, mechanistic interpretability), fairness metrics (demographic parity, equalised odds), robustness and adversarial attacks.

Tanglish (Tamil + English) Vilakkam
Explainable AI, fairness, adversarial robustness.

Learn more
YouTube search: explainable ai shap grad-cam

7. Robotics & embodied AI [Advanced]

Explanation (English)
Perception, planning, control, SLAM, simulation (Isaac, MuJoCo), imitation learning, vision-language-action models.

Tanglish (Tamil + English) Vilakkam
Robotics + AI.

Learn more
YouTube search: robot learning introduction

8. Knowledge graphs and semantic web [Advanced]

Explanation (English)
Entities and relations (RDF, Neo4j), graph embeddings, GraphRAG.

Tanglish (Tamil + English) Vilakkam
Knowledge graph, Neo4j, GraphRAG.

Example / Program
```
import networkx as nx
G=nx.DiGraph();G.add_edge('Chennai','TamilNadu',rel='in');print(list(G.edges(data=True)))
```

Learn more
YouTube search: Knowledge graphs and semantic web tutorial

9. Optimisation & operations research for ML (LP, MIP, genetic algorithms, simulated annealing) [Advanced]

Explanation (English)
Linear/integer programming, metaheuristics for scheduling, routing and hyper-parameter search (PuLP, OR-Tools, DEAP).

Tanglish (Tamil + English) Vilakkam
Optimization problems solve panna OR-Tools, PuLP.

Example / Program
```
from scipy.optimize import linprog
print(linprog([-1,-2],A_ub=[[1,1]],b_ub=[4],bounds=[(0,3),(0,3)]).x)
```

Learn more
YouTube search: Optimisation & operations research for ML (LP, MIP, genetic algorithms, simulated annealing) tutorial

10. Quantum machine learning (overview) [Advanced]

Explanation (English)
Quantum circuits as models (variational classifiers, kernels) using Qiskit / PennyLane. Mostly research-level today.

Tanglish (Tamil + English) Vilakkam
Quantum ML - research stage.

Learn more
YouTube search: quantum machine learning introduction

11. Neuro-symbolic AI, AGI, and AI safety / alignment [Advanced]

Explanation (English)
Combine neural nets with logic/search; study alignment, interpretability, scalable oversight and governance.

Tanglish (Tamil + English) Vilakkam
Neuro-symbolic, AI safety, alignment.

Learn more
YouTube search: ai alignment introduction

12. AI in domains: healthcare, finance, agriculture, education, manufacturing, NLP for Tamil [Intermediate]

Explanation (English)
Domain knowledge + data constraints + regulation decide success. Examples: credit scoring, medical imaging, crop disease detection, predictive maintenance, Tamil speech and OCR.

Tanglish (Tamil + English) Vilakkam
Domain-la AI apply: health, finance, agriculture, Tamil NLP.

Learn more
YouTube search: ai applications in agriculture healthcare finance

13. Reading and reproducing research papers [Advanced]

Explanation (English)
Read abstract -> figures -> method -> experiments; implement the core idea; use arXiv, Papers with Code alternatives (Hugging Face Papers), Connected Papers, Semantic Scholar.

Tanglish (Tamil + English) Vilakkam
Paper padikka: abstract, figures, method. Reproduce pannunga.

Learn more
YouTube search: how to read an ml research paper
arXiv cs.LG: https://arxiv.org/list/cs.LG/recent
HF Papers: https://huggingface.co/papers

14. Writing, portfolio, blogging and open-source contributions [Beginner]

Explanation (English)
Document projects with README, results, diagrams; publish on GitHub, Kaggle, Hugging Face Spaces; write blog posts; contribute to open-source issues labelled 'good first issue'.

Tanglish (Tamil + English) Vilakkam
GitHub portfolio, blog, open-source contribution.

Learn more
YouTube search: data science portfolio projects tips